Assignment 3

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1 Initial Value Problem

Solve the following initial value problem:

$$y'' - 2xy' - y = y^2 e^{-x^2}. \tag{1}$$

The initial conditions are

$$y(0) = 1, \qquad y'(0) = 0.$$

Use two steps of the Runge–Kutta method, as described in [1], with $h = 0.5$ and $h = 0.25$.

A C program was implemented to solve this initial value problem. The source code is available at `https://github.com/Julio-Medina/Runge-Kutta/tree/main/code`.

x	y	y'
0.00	1.00000	0.00000
0.50	1.28238	1.28208
1.00	2.50679	4.36570

Figure 1: RK4 result with $h = 0.5$.

x	y	y'
0.00	1.00000	0.00000
0.25	1.06447	0.53223
0.50	1.27572	1.20866

Figure 2: RK4 result with $h = 0.25$.

References

[1] Richard L. Burden and J. Douglas Faires, *Numerical Analysis*, 9th ed., Brooks/Cole, Cengage Learning. ISBN 978-0-538-73351-9.